

Project Report: DevOps Task Manager

Production-Ready Remote CI/CD Pipeline & AWS Infrastructure Management

1. Executive Summary

This report outlines the deployment architecture, configuration steps, and automated CI/CD pipeline for the **DevOps Task Manager** application. The project is designed to prioritize stability, resource efficiency, and simplicity:

- **Phase 1 — One-Time Server Setup:** Terraform provisions the AWS EC2 instance (`t3.micro` default) with Docker and k3s automatically installed via `user_data` (optimized with `--disable traefik --disable servicelb` to conserve memory).
- **Phase 2 — Automated CI/CD:** An automated Jenkins pipeline running locally on the developer's Windows machine checks out the source code, builds the Docker image, pushes it to Docker Hub, and executes a remote deployment via SSH.
- **Monitoring Architecture:** To operate stably within AWS free-tier limit constraints (1GB RAM on `t3.micro`), we adopt a **hybrid observability model**. Prometheus and Grafana run locally on the developer's machine using Docker Compose. Local Prometheus runs on port `9091` and Grafana on port `3000` , scraping metrics remotely from lightweight exporters (`node_exporter` and `kube-state-metrics`) running on the EC2 host.

By keeping the heavy build engine (Jenkins), query database (Prometheus), and visualization UI (Grafana) running locally, we keep the remote EC2 instance's memory footprint extremely low and guarantee high cluster stability.

2. Phase 1 — One-Time Server Setup

2.1 Host Specifications & Sizing Strategy

Running container orchestration and monitoring stacks on limited cloud resources requires careful memory budgeting. - **Instance Type Selection:** `t3.micro` (2 vCPUs, 1GB RAM) - *Sizing Strategy:* To operate within AWS free-tier limit constraints, we deploy on a single `t3.micro` instance. - *Optimization Strategy:* Running heavy Docker builds and Jenkins pipeline orchestration locally saves critical CPU and memory on the target server. Furthermore, the remote k3s cluster is optimized by disabling Traefik and ServiceLB, and the database metrics storage is offloaded to the local machine, preserving the EC2 instance's RAM for the core task-manager application.

2.2 Security Group Rules

The security group is configured to allow administration, API access, application routing, and remote metric collection:

Port	Protocol	Purpose	Access Source
22	TCP	SSH — Remote administration and Jenkins pipeline deployment	Restricted to Developer/Jenkins IP
80	TCP	HTTP — Task Manager application routing	Any (0.0.0.0/0)
6443	TCP	Kubernetes API — Remote kubectl	Restricted to Admin IP
9100	TCP	Node Exporter — Host system metrics	Restricted to Admin/Prometheus IP
30080	TCP	App NodePort — Exposes Task Manager frontend	Any (0.0.0.0/0)
30091	TCP	Kube-State-Metrics NodePort — Exposed cluster state metrics	Restricted to Admin/Prometheus IP

2.3 Auto-Installation via User Data

Terraform automatically bootstraps the instance using the `user_data` script, which installs Docker and installs an optimized k3s cluster:

```
curl -sL https://get.k3s.io | sh -s - \  
  --write-kubeconfig-mode 644 \  
  --node-external-ip=$(curl -s http://169.254.169.254/latest/meta-data/public-ipv4) \  
  --disable traefik \  
  --disable servicelb
```

2.4 Remote Exporters Setup (One-Time)

SSH into the EC2 instance and set up the lightweight exporters:

1. Setup Node Exporter (Host System Metrics)

```
wget https://github.com/prometheus/node_exporter/releases/download/v1.7.0/node_exporter-1.7.0.linux-amd64.tar.gz  
tar -xvf node_exporter-1.7.0.linux-amd64.tar.gz  
sudo mv node_exporter-1.7.0.linux-amd64/node_exporter /usr/local/bin/  
  
sudo tee /etc/systemd/system/node_exporter.service <<EOF  
[Unit]  
Description=Node Exporter  
After=network.target  
  
[Service]  
User=ubuntu  
ExecStart=/usr/local/bin/node_exporter  
  
[Install]  
WantedBy=multi-user.target  
EOF  
  
sudo systemctl daemon-reload  
sudo systemctl enable --now node_exporter
```

2. Setup Kube-State-Metrics (Cluster State Metrics)

The manifest file (`k8s/kube-state-metrics.yaml`) is copied and applied dynamically as part of the Kubernetes directory rollout. It exposes metrics on NodePort `30091` .

3. Phase 2 — Automated CI/CD

3.1 Repository Structure

```
devops-task-manager/  
├─ app/                # React frontend source code (Vite + React)  
├─ docker/             # Dockerfile for multi-stage image build  
├─ k8s/                # Kubernetes Deployment and Service manifests  
│   ├── deployment.yaml  
│   ├── service.yaml  
│   └─ kube-state-metrics.yaml  
├─ terraform/          # Terraform configurations  
│   ├── main.tf  
│   ├── variables.tf  
│   ├── outputs.tf  
│   └─ terraform.tfvars  
├─ jenkins/            # Jenkinsfile pipeline definition  
└─ monitoring/         # Local observability configurations  
    ├── README.md  
    ├── prometheus-local.yml  
    └─ docker-compose.yml
```

3.2 Automated CI/CD Workflow

The local Jenkins server coordinates the delivery pipeline:

```
sequenceDiagram
    autonumber
    actor Dev as Developer
    participant Git as GitHub
    participant Jen as Local Jenkins Server (Windows)
    participant DH as Docker Hub
    participant EC2 as AWS EC2 (k3s Node)

    Dev->>Git: git push
    Git->>Jen: Webhook Trigger
    Note over Jen: Git Checkout Source
    Jen->>Jen: Build Docker Image (Vite + React)
    Jen->>DH: Push Image with tag (komalpreet1703/task-manager:build-X)
    Jen->>Jen: Query EC2 IP dynamically via AWS CLI (PowerShell)
    Jen->>EC2: SCP updated k8s/ manifests using local PEM key
    Jen->>EC2: SSH: kubectl apply -f k8s/ using local PEM key
    Note over EC2: Pulls new image from Docker Hub & updates Pods
    Jen->>EC2: SSH: kubectl rollout status deployment/task-manager
    Jen-->>Dev: Pipeline Success
```

4. Implementation Artifacts

4.1 Dockerfile (`docker/Dockerfile`)

```
FROM node:20-alpine as build
WORKDIR /app
COPY app/package.json app/package-lock.json* ./
RUN npm install
COPY app/ ./
RUN npm run build

FROM nginx:alpine
RUN rm -rf /usr/share/nginx/html/*
COPY --from=build /app/dist /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

4.2 Jenkinsfile (jenkins/Jenkinsfile)

```

pipeline {
    agent any

    environment {
        DOCKER_HUB_CREDENTIALS_ID = 'dockerhub-creds'
        DOCKER_IMAGE = 'komalpreet1703/task-manager'
        IMAGE_TAG = "${env.BUILD_NUMBER}"

        PEM_FILE = 'C:\\jenkins-keys\\task-manager.pem'

        GIT_SSH = 'C:\\Program Files\\Git\\usr\\bin\\ssh.exe'
        GIT_SCP = 'C:\\Program Files\\Git\\usr\\bin\\scp.exe'
    }

    stages {
        stage('Checkout Source') {
            steps {
                checkout scm
            }
        }

        stage('Verify Tools') {
            steps {
                bat 'where aws'
                bat 'where ssh'
                bat 'where scp'
                bat '"%GIT_SSH%" -V'
                bat 'docker --version'
                bat 'kubectl version --client'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat """
                docker build ^
                -t ${DOCKER_IMAGE}:${IMAGE_TAG} ^
                -t ${DOCKER_IMAGE}:latest ^
                -f docker/Dockerfile .
                """
            }
        }

        stage('Push Docker Image') {
            steps {
                withCredentials([

```

```

        usernamePassword(
            credentialsId: DOCKER_HUB_CREDENTIALS_ID,
            usernameVariable: 'DOCKER_USER',
            passwordVariable: 'DOCKER_PASS'
        )
    ]) {
        bat """
        docker login -u %DOCKER_USER% -p %DOCKER_PASS%
        docker push ${DOCKER_IMAGE}:${IMAGE_TAG}
        docker push ${DOCKER_IMAGE}:latest
        """
    }
}

stage('Deploy to Kubernetes') {
    steps {
        script {
            def EC2_IP = powershell(
                script: """
                aws ec2 describe-instances `
                --region us-east-1 `
                --filters "Name=tag:Name,Values=Task-Manager-K3s-Node" "Name=instance-state-name,Values=running" `
                --query "Reservations[*].Instances[*].PublicIpAddress" `
                --output text
                """,
                returnStdout: true
            ).trim()

            echo "Deploying to EC2 IP: ${EC2_IP}"

            bat """
            "%GIT_SCP%" ^
            -o StrictHostKeyChecking=no ^
            -i "${PEM_FILE}" ^
            -r k8s ubuntu@${EC2_IP}:/home/ubuntu/
            """

            bat """
            "%GIT_SSH%" ^
            -o StrictHostKeyChecking=no ^
            -i "${PEM_FILE}" ^
            ubuntu@${EC2_IP} ^
            "sudo kubectl apply -f /home/ubuntu/k8s/ && sudo kubectl rollout status deploy"
            """
        }
    }
}

stage('Verify Services') {

```

```

steps {
  script {
    def EC2_IP = powershell(
      script: ""
      aws ec2 describe-instances `
      --region us-east-1 `
      --filters "Name=tag:Name,Values=Task-Manager-K3s-Node" "Name=instance-state-name,Values=running" `
      --query "Reservations[*].Instances[*].PublicIpAddress" `
      --output text
      "",
      returnStdout: true
    ).trim()

    bat ""
    "%GIT_SSH%" ^
    -o StrictHostKeyChecking=no ^
    -i "${PEM_FILE}" ^
    ubuntu@${EC2_IP} ^
    "sudo kubect1 get nodes && sudo kubect1 get pods -A && sudo kubect1 get svc -A"
    ""
  }
}

}

}

post {
  success {
    echo 'Deployment successful!'
  }
  failure {
    echo 'Pipeline failed!'
  }
  always {
    bat 'docker image prune -f'
  }
}
}

```


4.3 Deployment Manifesto (`k8s/deployment.yaml`)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: task-manager
  labels:
    app: task-manager
spec:
  replicas: 2
  selector:
    matchLabels:
      app: task-manager
  template:
    metadata:
      labels:
        app: task-manager
    spec:
      containers:
        - name: task-manager
          image: komalpreet1703/task-manager:latest
          imagePullPolicy: Always
          ports:
            - containerPort: 80
          resources:
            requests:
              memory: "64Mi"
              cpu: "50m"
            limits:
              memory: "128Mi"
              cpu: "100m"
```

4.4 Service Manifesto (`k8s/service.yaml`)

```
apiVersion: v1
kind: Service
metadata:
  name: task-manager-service
spec:
  type: NodePort
  selector:
    app: task-manager
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30080
```

4.5 Kube-State-Metrics Manifesto (`k8s/kube-state-metrics.yaml`)

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: kube-state-metrics
  namespace: kube-system
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: kube-state-metrics
rules:
- apiGroups: [""]
  resources:
    - configmaps
    - secrets
    - nodes
    - pods
    - services
    - resourcequotas
    - replicationcontrollers
    - limitranges
    - persistentvolumeclaims
    - persistentvolumes
    - namespaces
    - endpoints
  verbs: ["list", "watch"]
- apiGroups: ["apps"]
  resources:
    - statefulsets
    - daemonsets
    - deployments
    - replicaset
  verbs: ["list", "watch"]
- apiGroups: ["batch"]
  resources:
    - cronjobs
    - jobs
  verbs: ["list", "watch"]
- apiGroups: ["autoscaling"]
  resources:
    - horizontalpodautoscalers
  verbs: ["list", "watch"]
- apiGroups: ["authentication.k8s.io"]
  resources:
    - tokenreviews
  verbs: ["create"]
```

```

- apiGroups: ["authorization.k8s.io"]
  resources:
  - subjectaccessreviews
  verbs: ["create"]
- apiGroups: ["policy"]
  resources:
  - poddisruptionbudgets
  verbs: ["list", "watch"]
- apiGroups: ["certificates.k8s.io"]
  resources:
  - certificatesigningrequests
  verbs: ["list", "watch"]
- apiGroups: ["storage.k8s.io"]
  resources:
  - storageclasses
  - volumeattachments
  verbs: ["list", "watch"]
- apiGroups: ["admissionregistration.k8s.io"]
  resources:
  - mutatingwebhookconfigurations
  - validatingwebhookconfigurations
  verbs: ["list", "watch"]
- apiGroups: ["networking.k8s.io"]
  resources:
  - networkpolicies
  - ingresses
  verbs: ["list", "watch"]
- apiGroups: ["coordination.k8s.io"]
  resources:
  - leases
  verbs: ["list", "watch"]

```

```
---
```

```
apiVersion: rbac.authorization.k8s.io/v1
```

```
kind: ClusterRoleBinding
```

```
metadata:
```

```
  name: kube-state-metrics
```

```
roleRef:
```

```
  apiGroup: rbac.authorization.k8s.io
```

```
  kind: ClusterRole
```

```
  name: kube-state-metrics
```

```
subjects:
```

```
- kind: ServiceAccount
```

```
  name: kube-state-metrics
```

```
  namespace: kube-system
```

```
---
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: kube-state-metrics
```

```
  namespace: kube-system
```

```
  labels:
    app: kube-state-metrics
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kube-state-metrics
  template:
    metadata:
      labels:
        app: kube-state-metrics
    spec:
      serviceAccountName: kube-state-metrics
      containers:
        - name: kube-state-metrics
          image: registry.k8s.io/kube-state-metrics/kube-state-metrics:v2.10.0
          ports:
            - containerPort: 8080
              name: http-metrics
            - containerPort: 8081
              name: telemetry
          resources:
            requests:
              cpu: "10m"
              memory: "40Mi"
            limits:
              cpu: "50m"
              memory: "100Mi"
      ---
apiVersion: v1
kind: Service
metadata:
  name: kube-state-metrics
  namespace: kube-system
  labels:
    app: kube-state-metrics
spec:
  type: NodePort
  ports:
    - name: http-metrics
      port: 8080
      targetPort: 8080
      nodePort: 30091
  selector:
    app: kube-state-metrics
```

5. Observability Configurations

5.1 Local Prometheus Scrape Configuration (`monitoring/prometheus-local.yml`)

Local Prometheus is run via Docker Compose on the developer's machine on port `9091` and is configured to scrape the remote metrics endpoints directly over the internet:

```
global:
  scrape_interval: 15s
  evaluation_interval: 15s

scrape_configs:
  - job_name: 'remote-ec2-node'
    static_configs:
      - targets: ['44.195.19.206:9100']

  - job_name: 'remote-kube-state-metrics'
    static_configs:
      - targets: ['44.195.19.206:30091']

  - job_name: 'remote-task-manager-app'
    static_configs:
      - targets: ['44.195.19.206:30080']
```

5.2 Local Observability Compose Configuration (`monitoring/docker-compose.yml`)

```
version: '3.8'

services:
  prometheus:
    image: prom/prometheus:latest
    container_name: local-prometheus
    volumes:
      - ./prometheus-local.yml:/etc/prometheus/prometheus.yml
    ports:
      - "9091:9090"
    command:
      - "--config.file=/etc/prometheus/prometheus.yml"
      - "--storage.tsdb.retention.time=15d"
    restart: unless-stopped

  grafana:
    image: grafana/grafana:latest
    container_name: local-grafana
    ports:
      - "3000:3000"
    environment:
      - GF_SECURITY_ADMIN_PASSWORD=admin123
    restart: unless-stopped
```

6. Practical Windows Jenkins Deployment Challenges

Running a remote Linux-targeting pipeline from a Windows Jenkins environment presents several unique engineering challenges that were solved in this architecture:

1. **Path Resolution & Command Inheritance:** Windows command execution does not inherit the user-level path variable configuration in the same manner as Linux. To bypass OpenSSH path execution errors and ensure commands use the correct tools, we explicitly locate and declare variables for the Git Bash bundle binaries:
2. `GIT_SSH = 'C:\\Program Files\\Git\\usr\\bin\\ssh.exe'`
3. `GIT_SCP = 'C:\\Program Files\\Git\\usr\\bin\\scp.exe'` Using these direct references avoids issues where Jenkins cannot resolve shell executable wrappers or uses the wrong system SSH client.

4. **Private Key Permissions (`task-manager.pem`):** In Windows environments, checking out private keys or utilizing Jenkins credential wrappers often triggers key permission errors (`permissions are too open` or access denial). By storing the private key locally at a secure location (`C:\jenkins-keys\task-manager.pem`) and explicitly referencing this path, the Windows ssh/scp binary loads the key file securely and directly.
 5. **Multi-line Command Continuations:** Windows command processor (`cmd.exe`) uses the caret symbol (`^`) as the multi-line continuation character rather than the Unix backslash (`\`). Docker commands are formatted accordingly inside triple-quoted `bat` steps to ensure correct compilation.
-

7. Conclusion

Separating the DevOps workflow into a one-time server bootstrap phase and a dynamic local CI/CD deployment phase maximizes deployment stability. By offloading metrics collection (Prometheus) and dashboarding (Grafana) to the local developer environment on ports `9091` and `3000`, the remote `t3.micro` EC2 instance stays lightweight and operates efficiently within free-tier resource limits. This hybrid architecture simulates a professional, production-ready DevOps infrastructure pipeline.